H-BIOT: Hierarchical Transformer for Automated MEG Spike Detection

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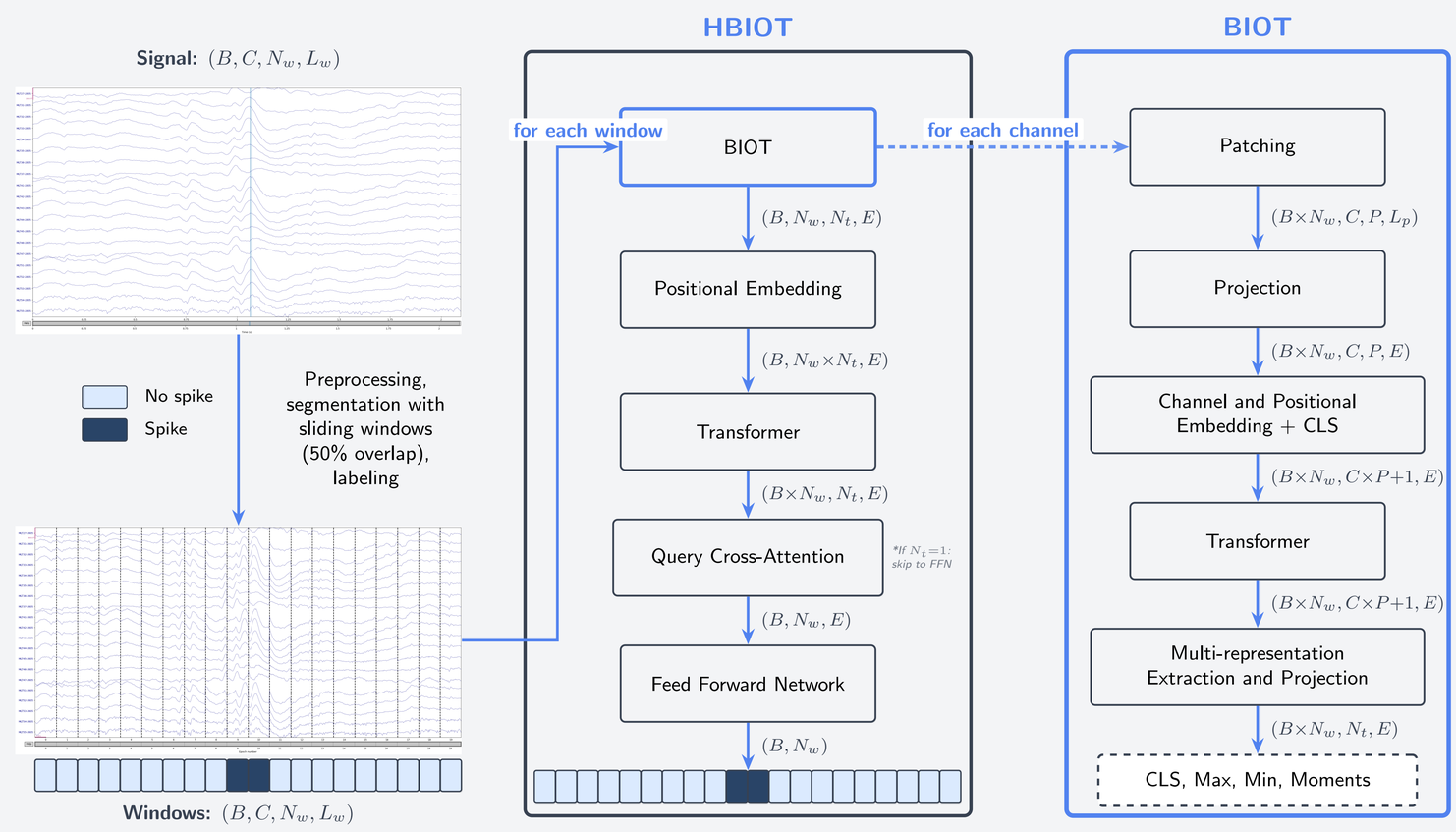
H-BIOT Architecture



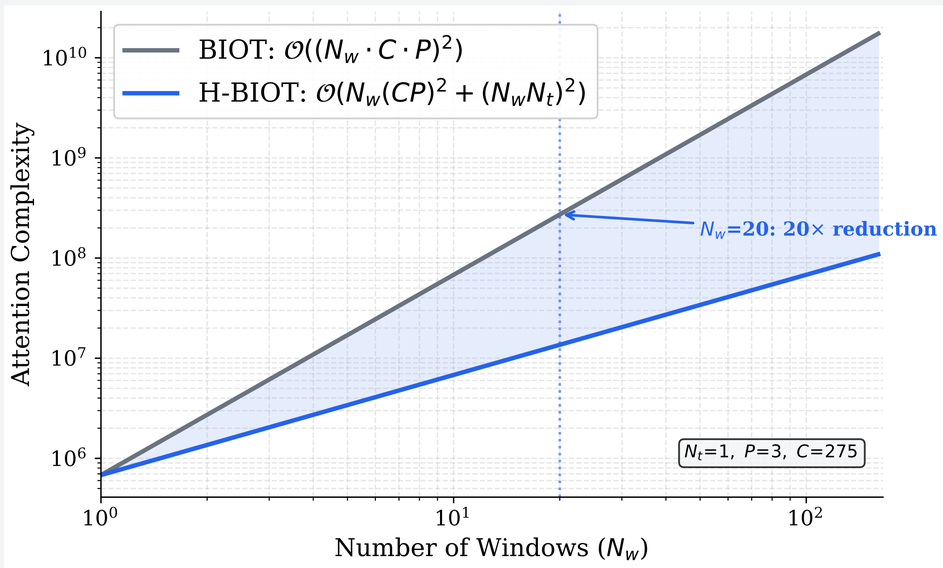
B: batch size, N_w: # windows, C: # channels, E: embedding size, P: patches per window, N_t: tokens per window, L_w: window length, L_p: patch length.

Key Improvements: Two-Stage Window Factorization

Stage 1 - Intra-window attention: Each 200 ms window is processed independently by a BIOT encoder (4 layers, 4 heads, FlashAttention) as a sequence of 275 channels × 3 patches = 825 tokens.

Stage 2 - Inter-window attention: Window-level representations are arranged into a new sequence and processed by a second transformer with learnable positional embeddings to capture **extended temporal context** for joint temporal modelling.

Complexity reduction (for $N_t \ll N_w \ll C$, e.g. $N_t = 1$, $P = 3$, $C = 275$):



Introduction & Motivation

Epilepsy surgery requires accurate localization of the epileptogenic zone. **Interictal spikes** in magnetoencephalography (MEG) are key biomarkers for this localization [1].

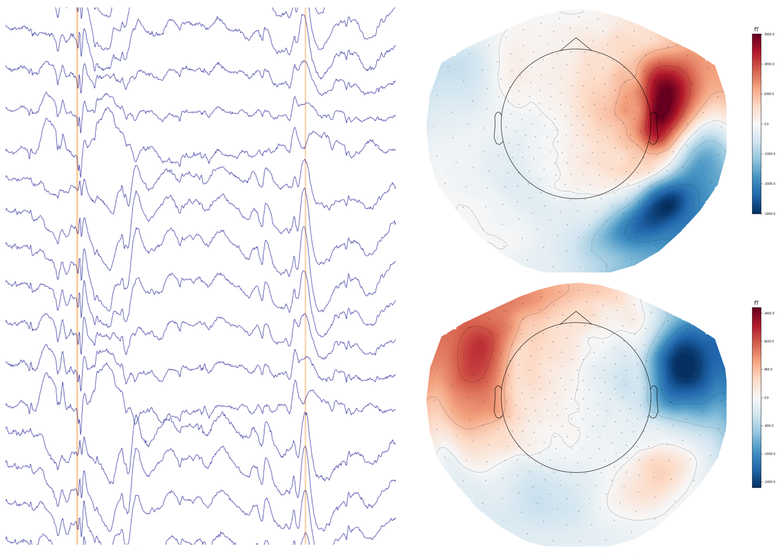
Challenges of manual annotation:

- Time-consuming: long continuous recordings (275 channels)
- Inter-rater agreement only fair to moderate [2,3]
- No standardized visual interpretation guidelines for MEG

Current landscape: Latest automated methods rely mainly on CNNs. Current adaptations of **existing transformer architectures fail to achieve state-of-the-art** performances in MEG.

What Is an Interictal Spike?

An interictal epileptic spike is a **brief transient waveform** (<100 ms) reflecting abnormal neuronal discharges between seizures, characterized by a sharp morphology and high amplitude relative to background activity.



MEG recording with two annotated interictal spikes (left).
Topographic maps show magnetic field distribution at spike locations (right).

Dataset & Experimental Setup

229 recordings **78 patients** 275-channel CTF MEG 3-min recordings	**5,553 spikes** (median: 6/min) **2.7% positive** severe imbalance	**Patient-level** split, stratified on spike count Train: 54, Val: 14 **Test: 10 patients**

- Preprocessing: 200 Hz downsampling, 1-99 Hz bandpass, 50 Hz notch, per-recording z-score normalization.
- Segmentation: 200 ms windows, 50% overlap.
- Evaluation: 5-fold patient-stratified CV. Results are reported as mean ± std on the total window count of the test set.

Training Strategy

- Loss: Focal loss ($\gamma = 2$, $\alpha = 0.25$) for class imbalance
- Augmentation: Gaussian noise ($\sigma = 0.1$), random channel masking, dynamic sequence sampling
- Optimization: AdamW (lr= 3×10^{-4}), ReduceLROnPlateau, early stopping on PR-AUC
- Threshold: F1-optimized per fold on validation set
- Framework & hardware: Lightning, FP16 mixed precision, H100 gpu

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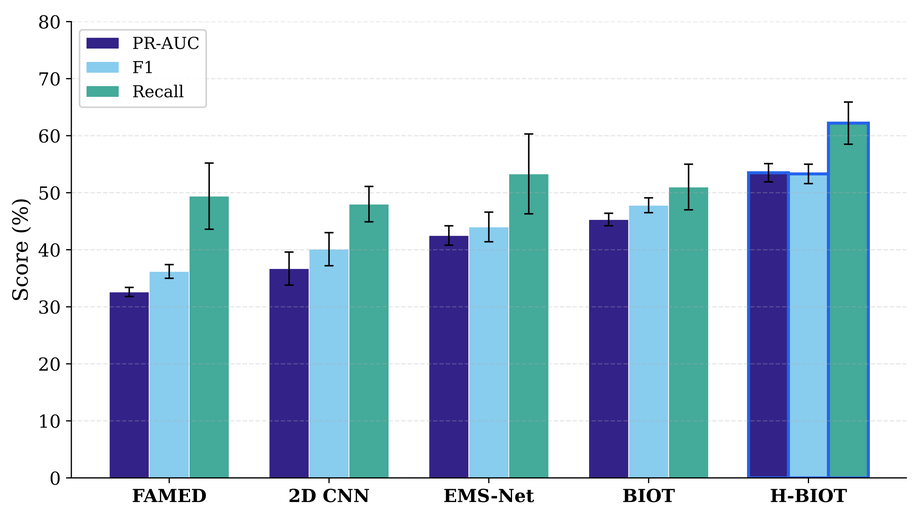
Code
github.com/Malchemis/HBIOT

Results: Model Comparison

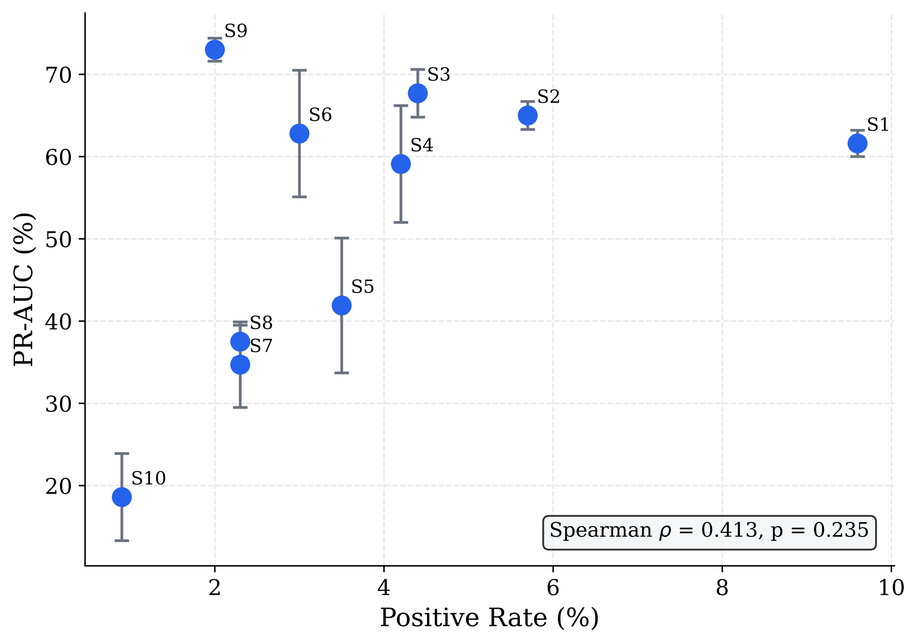


Model	PR-AUC	F1	Prec.	Recall	Spec.	ROC-AUC
FAMED [5]	32.6±0.8	36.2±1.2	29.2±3.9	49.4±5.8	95.1±1.3	86.8±0.3
2D CNN [6]	36.7±2.9	40.1±2.9	34.8±4.9	48.0±3.1	96.4±0.8	87.3±1.6
EMS-Net [7]	42.5±1.7	44.0±2.6	38.7±5.9	53.3±7.0	96.4±1.7	89.0±0.8
BIOT [8]	45.3±1.1	47.8±1.3	45.5±4.0	51.0±4.0	97.6±0.6	91.2±0.4
H-BIOT	**53.5****±1.6	**53.3***±1.7	47.1±4.5	**62.2**±3.7	97.2±0.6	**94.9****±0.6

$^{*}p < 0.05$, $^{**}p < 0.01$ (Holm-corrected paired t-test, H-BIOT vs each baseline)

+18% relative improvement in PR-AUC ($p < 0.01$) and **+4 pp ROC-AUC** ($p < 0.01$) over all baselines.
Recall gains (+11.2 pp vs BIOT) with comparable **Specificity** (97.2%).

Subject-Level Variability

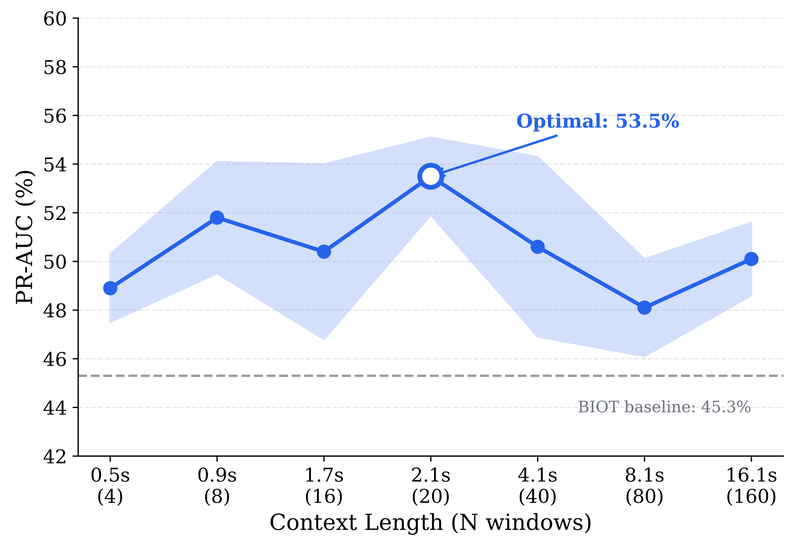


Large variations: PR-AUC from 18.6% to 73.0%, having a weak correlation ($\rho = 0.413$, $p = 0.235$) with positive rate. **The spike rate fails to explain subject variability**. Intrinsic spike characteristics (morphology, SNR, spatial extent) dominate over quantity.

Ablation: Temporal Context Length

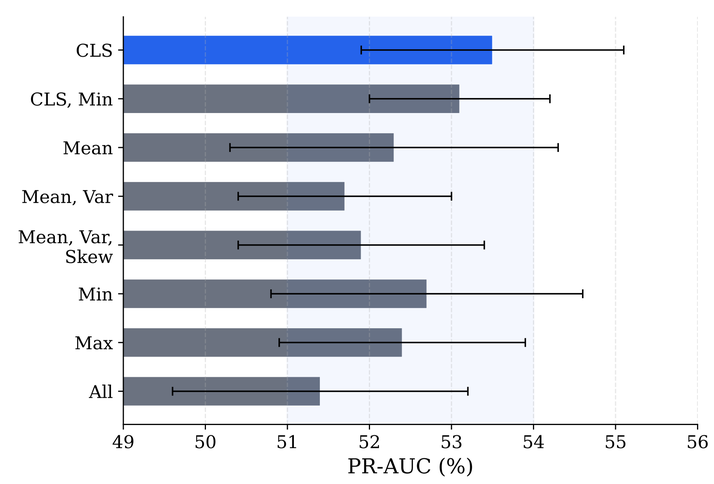
Key finding: Performance peaks at **2.1 seconds** (20 windows). Only the shortest context (0.5 s) is significantly worse ($p < 0.01$ on PR-AUC); other lengths are statistically comparable.

Interpretation: A minimum of ∼1 s of context is needed to capture the spike and surrounding network state (pre-spike build-up, post-spike slow wave) that disambiguates true spikes from artifacts. Beyond this threshold, performance is robust to context length.



⇒ Spikes may be better characterized by **network dynamics** rather than isolated waveforms.

Ablation: Token Representations



Key finding: Single-token representations (CLS, Mean, Min, Max) perform **equivalently to multi-token combinations** ; no significant differences on any metric (Holm-corrected). Intra-window self-attention already aggregates spatial-temporal features effectively. Simpler architectures with $N_t = 1$ suffice.

Discussion & Realistic Assessment

H-BIOT achieves **statistically significant improvements** in ranking metrics (PR-AUC, ROC-AUC; $p < 0.01$) over all baselines. Our F1 of 53.3% contrasts with >90% in some literature, but **since inter-rater agreement is only fair to moderate** [2,3], near-perfect metrics may not reflect realistic clinical performance.

Limitations & Future Work:

- Single-center → multi-center evaluation
- Window-level → event-level detection with onset timing
- Semi-supervised pre-training on unlabeled MEG
- Few-shot adaptation to patients and analysis of failure modes

References

[1] J. Jung, R. Bouet, C. Delpuech, et al., "The value of magnetoencephalography for seizure-onset zone localization in MRI-negative partial epilepsy," *Brain*, vol. 136, no. 12, pp. 3176-3186, 2013.
[2] J. Jing, A. Herlopian, I. Karakis, et al., "Interrater reliability of experts in identifying interictal epileptiform discharges in electroencephalograms," *JAMA Neurology*, vol. 77, no. 1, pp. 49-57, 2020.
[3] J.J. Halford, R.J. Schalkoff, J. Zhou, et al., "Standardized database development for EEG epileptiform transient detection," *J. Neuroscience Methods*, vol. 212, no. 2, pp. 308-316, 2013.
[4] M. de Curtis and G. Avanzini, "Interictal spikes in focal epileptogenesis," *Prog. Neurobiol.*, vol. 63, no. 5, pp. 541-567, 2001.
[5] R. Hirano, T. Emura, O. Nakata, et al., "Fully-automated spike detection and dipole analysis of epileptic MEG using deep learning," *IEEE Trans. Med. Imaging*, vol. 41, no. 10, pp. 2879-2890, 2022.
[6] P. Mouches, J. Jung, A. Demasson, et al., "Automated interictal epileptic spike detection from simple and noisy annotations in MEG data," 2025.
[7] L. Zheng, P. Liao, S. Luo, et al., "EMS-Net: A deep learning method for autodetecting epileptic magnetoencephalography spikes," *IEEE Trans. Med. Imaging*, vol. 39, no. 6, pp. 1833-1844, 2020.
[8] C. Yang, M. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," in *Advances in Neural Information Processing Systems*, vol. 36, pp. 78240-78260, 2023.